

Qn	Solution
1(a)	Liquid pressure is directly proportional to depth of liquid. The pressure at the bottom of the block is higher than the pressure at the top of the block. Since pressure is force per unit area, the upward force at the bottom is larger than the downwards force at the top. The resultant force by liquid on the block is upwards. This force is known as the upthrust.
1(b)(i)	$U = \text{weight of fluid displaced}$ $= mg$ $= pVg$ $= (1.0 \times 10^3)(27.8 \times 10^{-6})(9.81) = 0.27272 \approx 0.27 \text{ N}$
1(b)(ii)	Taking moment about pivot, anti-clockwise moment = clockwise moment $W(8.3) = F(18.0) \dots\dots\dots(1)$ $W(7.8) + (0.27272)(19.0) = F(19.0) \dots\dots\dots(2)$ Solving, $F = 2.4860 \approx 2.5 \text{ N}$
2(a)	At maximum height, kinetic energy at the top will be minimum